

The world keeps moving while agents think.

RT-Safe Benchmark

Task: navigation to the goal safely

01 / THE REAL-TIME PROBLEM

Safety includes the time spent deciding.

Observe street views, recent motion, and the navigation goal.



Static evaluation

INFERENCE World paused	ACTION World moves
---------------------------	-----------------------

RT-SAFE · real-time

INFERENCE World moves	ACTION World moves
--------------------------	-----------------------

An observation can become stale.
Collisions can occur before an action starts.

What changes when we evaluate safety throughout the episode?

RQ1 / EIGHT VISION-LANGUAGE MODELS

Arrival rarely means safe arrival.

0.7% finish without a safety event

Despite 94.4% task success.

Task success

Safe success

0.7%

94.4%

Episodes (%)

Behavior matters.

Inkling uses longer moves and fewer decisions; Grok has slower responses and more collisions.

Inkling

Grok

38.5 decisions · 21.9 collisions

60.9 decisions · 59.5 collisions

All difficulties · 108 episodes per model.

Provider-default reasoning · Counts per episode.

RQ2 / STATIC → REAL-TIME

Pausing hides safety failures.

12.3× more collisions

Safe completion: 19.8% → 0.7%.

Static

Real-time

3.3

40.7

Collisions per episode

About 84% of real-time collisions occur during inference.

Task success stays high: 91.3% → 94.1%.

Hard · 288 matched model-route pairs · Default effort.

Timing instructions also differ between modes.

RQ3 / DEFAULT → HIGHER REASONING

More thinking adds exposure.

40.7 → 62.0

Mean collisions per episode.

Default

Higher

40.7

62.0

Collisions per episode

Action collisions fall: 6.4 → 5.9.

Inference collisions rise: 34.3 → 56.1.

Safer execution can coexist with worse overall safety.

Hard, real-time · 288 episodes per condition.

Effort levels are provider-specific.

Evaluate decision quality and decision latency together.

Source: RT-SAFE paper §§2–3.4, Figures 2–5 and Appendix B.
Values rounded as reported. Project: [xtrose9.github.io/RT-SAFE/](https://github.com/xtrose9/RT-SAFE/)